

A Direct Proof of the Heisenberg Singular-Vector Factorization

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August 11, 2026

Source problem and answer

Křížka and Somberg study the Heisenberg parabolic of $\mathfrak{sl}(n+2, \mathbb{C})$. In their notation the opposite nilradical $\bar{\mathfrak{u}}$ has basis $f_1, \dots, f_n, g_1, \dots, g_n, c$, with

$$[f_i, g_j] = -\delta_{ij}c, \quad c \text{ central.}$$

On p. 35, after verifying the cases $a = 1, 2$, they conjecture that whenever

$$\lambda = \lambda_1\omega_1 + \lambda_2\omega_{n+1}, \quad \mu = (\lambda_1 - a)\omega_1 + (\lambda_2 - a)\omega_{n+1}, \quad \lambda_1 + \lambda_2 + n = a - 1,$$

the scalar singular vector factors as

$$\varphi(v_\mu) = \prod_{j=0}^{a-1} \left(\sum_{i=1}^n f_i g_i + \frac{\lambda_1 - \lambda_2 + n - a + 2j + 1}{2} c \right) v_\lambda. \quad (\text{F})$$

They write that they do not know a direct proof. The answer is affirmative. The proof below derives (F) directly from their polynomial formula for the singular vector and the Weyl-to-normal ordering transform.

The examples (1) and (2) suggest the following remarkable conjectural factorization property of singular vectors $\varphi(v_\mu) \in M_{\mathfrak{p}}^{\mathfrak{g}}(\lambda)$ inducing homomorphisms of scalar generalized Verma modules, see Theorem 2.4 regarded by (2.18) as elements of the (non-commutative) universal enveloping algebra $U(\bar{\mathfrak{u}})$. For all $\lambda = \lambda_1\omega_1 + \lambda_2\omega_{n+1}$, $\mu = (\lambda_1 - a)\omega_1 + (\lambda_2 - a)\omega_{n+1}$, $\lambda_1 + \lambda_2 + n = a - 1$ and $a \in \mathbb{N}$, we conjecture to hold

$$\varphi(v_\mu) = \prod_{j=0}^{a-1} \left(\sum_{i=1}^n f_i g_i + \frac{1}{2}(\lambda_1 - \lambda_2 + n - a + 2j + 1)c \right) v_\lambda. \quad (3.140)$$

We do not know a direct proof of this observation.

Proof intuition

Put $A = \sum_i f_i g_i$. The weight relation makes the j th conjectured factor simply $A + (j - \lambda_2)c$, so the right side of (F) is a rising factorial in $A/c - \lambda_2$.

The source constructs the same singular vector on the commutative Fourier side as $T_{a,a}^\lambda(q^a)$, where $q = \sum_i x_i y_i$ and the coefficients of T have an explicit generating function. The only apparent difficulty is converting this commutative polynomial back through symmetrization. Exponential generating functions remove it: the exact symmetrization of e^{uq} is

$$\beta(e^{uq}) = (1 - cu/2)^{-n} \left(\frac{1 + cu/2}{1 - cu/2} \right)^{A/c}.$$

After inserting the source coefficient generating function, the desired vector becomes one coefficient of a two-binomial product. A fractional linear change of variable turns that coefficient into $\binom{A/c - \lambda_2 + a - 1}{a}$, which is precisely the conjectured product.

Statement

Theorem 1 (Factorization conjecture (3.140)). *Let $n > 2$, $a \in \mathbb{N}$, and $\lambda_1 + \lambda_2 + n = a - 1$. Under the conventions and the normalization of Křížka–Somberg, the singular vector represented on the Fourier side by*

$$P_\lambda = T_{a,a}^\lambda(q^a)$$

has the exact PBW factorization

$$\varphi(v_\mu) = \prod_{j=0}^{a-1} (A + (j - \lambda_2)c) v_\lambda, \quad A = \sum_{i=1}^n f_i g_i.$$

Equivalently, formula (F) holds for every parameter allowed by the source statement.

Proof

We use formal power series only as coefficient-extraction devices; every coefficient used below is a polynomial identity in A and c .

1. The source polynomial and the Fourier sign

Let $\beta : S(\bar{\mathbf{u}}) \rightarrow U(\bar{\mathbf{u}})$ be symmetrization. Theorem 2.4 and the algebraic Fourier transform of the source imply, for a homogeneous commutative monomial p of ordinary degree d ,

$$(\tau \circ \Phi_{\lambda+\rho})(\beta(p)v_\lambda) = (-1)^d p.$$

The source defines

$$T_{a,a}^\lambda = \sum_{k \geq 0} \frac{\alpha_k}{2^k} (z \partial_q)^k, \quad g(w) := \sum_{k \geq 0} \alpha_k w^k = (1+w)^{a-\lambda_1-1} (1-w)^{a-\lambda_2-1}.$$

Writing $(a)^{\underline{k}} = a!/(a-k)!$, we therefore obtain the normalized preimage of P_λ as

$$V_a = \beta \left(\sum_{k=0}^a \frac{(-1)^k \alpha_k}{2^k} (a)^{\underline{k}} c^k q^{a-k} \right) v_\lambda. \quad (1)$$

Indeed, $z^k q^{a-k}$ has ordinary degree $2a - k$, hence Fourier sign $(-1)^k$. The leading term in (1) is normalized to A^a , so no undetermined scalar remains.

The weight relation gives

$$g(w) = (1+w)^{\lambda_2+n} (1-w)^{\lambda_1+n}. \quad (2)$$

2. Exact symmetrization of the radial exponential

Let N denote normal ordering: all f_i are put to the left of all g_i . Set $q = \sum_i f_i g_i$ in the symmetric algebra and $\Delta = \sum_i \partial_{f_i} \partial_{g_i}$. The central commutator and the Baker–Campbell–Hausdorff formula give the standard ordering identity

$$N^{-1}\beta(p) = \exp\left(\frac{c}{2}\Delta\right)p. \quad (3)$$

For completeness, (3) follows by applying both sides to $e^{\sum_i (\xi_i f_i + \eta_i g_i)}$: the discrepancy between the Weyl and normal exponentials is $e^{(c/2)\sum_i \xi_i \eta_i}$.

Solving the elementary heat equation produced by (3) gives

$$\exp\left(\frac{c}{2}\Delta\right)e^{uq} = (1 - cu/2)^{-n} \exp\left(\frac{u}{1 - cu/2}q\right).$$

For $A_i = f_i g_i$, the relations imply

$$f_i^m g_i^m = \prod_{r=0}^{m-1} (A_i - rc), \quad N(e^{vq}) = \prod_i (1 + cv)^{A_i/c} = (1 + cv)^{A/c}.$$

Consequently

$$\boxed{\beta(e^{uq}) = (1 - cu/2)^{-n} \left(\frac{1 + cu/2}{1 - cu/2}\right)^{A/c}.} \quad (4)$$

The temporary localization at c in the exponent is harmless: each coefficient of (4), and the final identity, lies in $\mathbb{C}[A, c]$.

3. Coefficient extraction

The expression inside β in (1) is

$$a![u^a], g(-cu/2)e^{uq}.$$

Combining (2) and (4) therefore yields

$$V_a/v_\lambda = a![u^a](1 - cu/2)^{\lambda_2 - A/c}(1 + cu/2)^{\lambda_1 + n + A/c}. \quad (5)$$

Put

$$X = A/c, \quad Y = X - \lambda_2, \quad t = cu.$$

Since $\lambda_1 + n = a - 1 - \lambda_2$, equation (5) becomes

$$V_a/v_\lambda = c^a a![t^a](1 - t/2)^{-Y}(1 + t/2)^{a-1+Y}. \quad (6)$$

It remains to evaluate one universal coefficient. Make the formal change

$$v = \frac{t}{1 - t/2}, \quad t = \frac{v}{1 + v/2}.$$

Then

$$(1 - t/2)^{-Y}(1 + t/2)^{a-1+Y} = (1 + v)^{a-1+Y}(1 + v/2)^{-(a-1)}, \quad \frac{dt}{t^{a+1}} = (1 + v/2)^{a-1} \frac{dv}{v^{a+1}}.$$

Taking formal residues, the two extra factors cancel and give

$$[t^a](1 - t/2)^{-Y}(1 + t/2)^{a-1+Y} = [v^a](1 + v)^{a-1+Y} = \binom{a-1+Y}{a} = \frac{(Y)_a}{a!}. \quad (7)$$

Equations (6)–(7) now imply

$$V_a/v_\lambda = c^a(Y)_a = \prod_{j=0}^{a-1} (A + (j - \lambda_2)c). \quad (8)$$

Finally, the source coefficient simplifies as

$$\frac{\lambda_1 - \lambda_2 + n - a + 2j + 1}{2} = j - \lambda_2$$

under $a = \lambda_1 + \lambda_2 + n + 1$. Thus (8) is exactly (F), completing the proof. $\square$

Verification notes

The script `code/verify_factorization.py` performs two independent exact checks. First it verifies the universal coefficient identity (7) symbolically for $1 \leq a \leq 10$. Second, for several rational parameter choices, it reconstructs the source coefficients α_k , applies (4) term by term to (1), and compares the result with the product in (8). All checks pass over exact rational polynomial arithmetic.

The most sensitive review points are the signs. They are fixed by $[f_i, g_i] = -c$, which gives $\beta(f_i g_i) = f_i g_i + c/2$, and by the fact that $z^k q^{a-k}$ has ordinary degree $2a - k$ under Theorem 2.4, producing the $(-1)^k$ in (1). The formulas reproduce the source’s displayed cases $a = 1$ and $a = 2$.

Novelty boundary and human-review recommendation

A bounded search through August 11, 2026 used arXiv id 1502.07095, the exact title and authors, exact phrases around “direct proof” and “factorization property,” and formula/core-keyword variants. The run’s cheap indexes were also searched. The arXiv source and the authors’ public bibliography were checked; a nearby later paper by Pandžić and Somberg on $\mathfrak{sl}_2$ -actions was inspected and treats different commutative-nilradical examples. No later proof of (3.140) was located. This is a bounded novelty check, not a guarantee of priority.

Recommendation: review as a likely valid full solution. A human verifier should focus first on the source-to-(1) Fourier sign and then on the normal ordering identity (3); after those two conventions are confirmed, the rest is a formal coefficient identity.

References

- [1] L. Křížka and P. Somberg, *Algebraic analysis on scalar generalized Verma modules of Heisenberg parabolic type I.: A_n -series*, Transform. Groups 22 (2017), 403–451; arXiv:1502.07095.
- [2] P. Pandžić and P. Somberg, *Branching problems and $\mathfrak{sl}(2, \mathbb{C})$ -actions*, Archivum Math. (Brno) 51 (2015), 331–346; arXiv:1511.03048.